

Program : Diploma in Information Technology	
Course Code : 3266	Course Title: Computer System Administration Lab I
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide hands-on experience with the concept of Linux and Linux commands.
- To provide practical knowledge with installation of Linux operating systems.
- To provide basic concepts in shell script and develop programming skills .

Course Pre-requisites:

Topic	Course Code	Course Title	Semester
Basic Knowledge of IT systems		Introduction to IT system	1
Control structure and loop		Problem Solving & Programming	2

Course Outcomes:

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Apply basic concepts of Linux operating system and experiment with any two Linux operating systems.	10	Applying
CO2	Apply basic Linux commands	10	Applying
CO3	Apply basic concepts of simple Shell Scripting	10	Applying
CO4	Develop Shell Scripting using conditional and loop control structure	12	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3		3				
CO3	3		3				
CO4	3		3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Apply basic Concepts of Linux Operating System and experiment with any two Linux Operating Systems.		
M1.01	Experiment with installation of Ubuntu operating system	3	Applying
M1.02	Experiment with installation of fedora	3	Applying
M1.03	Experiment with installation of Red Hat enterprise	4	Applying
CO2	Apply Basic Linux Commands.		
M2.01	Make use of directory management commands: ls, cd, pwd, mkdir, rmdir	2	Applying
M2.02	Make use of File Management commands: cat, chmod, cp, mv, rm, more.	3	Applying
M2.03	Experiment with the general-purpose commands: wc, od, lp, cal, date, who, tty, ln	3	Applying
M2.04	Experiment with the simple filters: pr, head, tail, cut, paste, nl, sort	2	Applying
	Lab Exam – I	1.5	
CO3	Apply basic Concepts of Simple Shell Scripting		
M3.01	Make use of variables in shell script	3	Applying
M3.02	Make use of input and output commands in shell script	3	Applying
M3.03	Experiment with the concept of basic operators and shell script	4	Applying
CO4	Develop Shell Scripting using conditional and loop control structure.		

M4.01	Experiment with the concept of decision making and looping structure in shell script to solve simple problems	6	Applying
M4.02	Develop programs using Loop controls in shell scripts.	6	Applying
	Lab Exam – II	1.5	

*****Suggested Open-ended Experiments:**

Students can do open-ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation and a score of 15 marks. Students should prepare a separate report on open ended experiment of their choice.

Example: Boot your computer and

1. Install Red Hat Enterprise Linux
2. Trouble shooting.

Text / Reference:

T/R	Book Title / Author
T1	Richard Blum and Christine Bresnahan, “Linux command line and shell scripting Bible”, Second Edition
T2	Steve Parker, “Shell Scripting Expert Recipes for Linux, Bash and More”, Wiley India Edition

Online Resources:

Sl.No	Website Link
1	https://www.linux.com/what-is-linux/
2	https://www.redhat.com/en/topics/linux/what-is-linux
3	https://www.guru99.com/must-know-linux-commands.htm